



Hybrid impedance and admittance control of robot manipulator with unknown environment

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Abstract

This paper presents a novel impedance control strategy to improve the performance of a robot manipulator. Impedance control and admittance control have complementary effects on the stability and performance of a control system. Impedance control works well in stiff environments, whereas admittance control works well in soft environments. In this paper, we propose a hybrid impedance and admittance control strategy that switches the controller based on the switching condition. If proper switching between impedance control and admittance control is achieved, the controller will have the advantages of both the control strategies. The proposed schemes were evaluated through simulations using a 2-DOF manipulator. Experiments were conducted using an actual robot. The results of the simulation and experiments performed confirmed that the proposed control strategy improves the performance of the robot manipulator.

Keywords Impedance control · Admittance control · Force control · Sensor-based control

1 Introduction

Impedance control of a robot when it interacts with its environment is an actively researched control method. Impedance control is an indirect force control method that involves dynamic interactions between the position and force of the end-effector. This enables the manipulator to be controlled in free space and perform motions that are compliant with the environment.

Impedance control can be implemented in two ways, namely “impedance control” and “admittance control” [1]. Impedance control, which is also called force-based impedance control, is a control system that takes displacement as the input and provides force as the output. In contrast, admittance control, also called position-based impedance control, is a control system that takes force as the input and provides displacement as the output.

Impedance control outputs the force or torque for the desired motion based on the calculated dynamic models of an actual manipulator. This enables the manipulator to have stable dynamic interactions with stiff environments. Ideally, low-friction joints and direct-drive actuators enable a manipulator to interact with the environment with accurate force and position; for example, a phantom haptic device from 3D Systems can interact with a human hand [2]. However, the position accuracy depends on the backdrivability and friction of the manipulator. For the systems with poor backdrivability and friction, such as a conventional robot system, these drawbacks of impedance control can be overcome by using inner-loop torque sensing/control, such as that performed in the lightweight robots from DLR [3] and the quadruped robot AiDIN VI [4–6]. However, torque sensors are expensive and difficult to implement in conventional robotic systems. Without a torque sensor, the position accuracy can be improved by a high desired stiffness and damping of impedance. However, with a low desired stiffness, the Impedance Control still has poor position accuracy during free motion [7]. The errors of dynamic models can also degrade the position accuracy. As a result, impedance control is advantageous for compliant behavior in a stiff environments.

Unlike impedance control, admittance control outputs the desired position to the position controller of the system. This provides a high position accuracy during free motion. How-

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ever, it makes the system unstable during its interaction with stiff environments. To overcome the problem of instability in stiff environments, passive compliant components such as series elastic actuation [8],[9] or soft end-effectors [10] are used. However, passive compliant components degrade the performance of admittance control. The performance and stability of admittance control mainly depend on the position controller, which effectively compensates the friction of the joints [11]. Therefore, admittance control is advantageous for stiff behavior in a soft environments compared to compliant behavior.

Impedance control and admittance control have complementary characteristics. Both impedance control and admittance control exhibit insufficient performance in environments with large changes in stiffness. Studies been conducted on improving the performance of impedance control and admittance control by varying the impedance parameters. In several studies on the interaction of a robot with a human, the human intention was estimated using sensors and state information, and the impedance was adjusted continuously [12–14]. In some studies, the human arm impedance was estimated to operate a variable impedance controller [15,16]. Roveda et al. estimated the stiffness of the environment to improve the impedance control during interaction with unknown environments [17,18]. However, the variable impedance control strategies used in these studies focused on performing specific tasks rather than general tasks. Furthermore, when the stiffness of the environment changes, impedance control and admittance control should be properly selected to achieve the best performance.

To obtain the best controller for any given task, Ott et al. proposed a hybrid framework for unified impedance and admittance control, in which the controllers were rapidly switched [19,20]. By combining both impedance and admittance controls, the overall performance of the control system was improved. They introduced δ and n gains for the duty cycle of the switching system and interpolated the response between the impedance control and admittance control. However, it is necessary to properly set the duty cycle based on the stiffness of the environment. Based on the unified impedance and admittance control framework, Cavenago et al. proposed an adaptive hybrid framework in which the n gain was changed using a neural network [21]. These approaches exhibited good performance in environments with varying stiffness values. However, in a real-life environment, the stiffness of the environment is typically very soft or very hard. For example, in the case of a deburring task, one direction has a very stiff environment, whereas the other direction has free space. Therefore, a new strategy is required to address such complicated tasks.

In this study, a hybrid impedance and admittance control strategy is proposed. The basic scheme is similar to that of the unified impedance and admittance scheme. However,

the switching condition is different. To achieve stability of the switching system, the switching controller of the unified impedance and admittance control is adopted. With regard to the switching condition, a guided trajectory $\mathbf{x}_g$ and an acceptable position error α are newly introduced. When the position error $\|\mathbf{x}_g - \mathbf{x}\|$ is larger than α , the impedance control is switched to admittance control. To ensure the stability of the admittance control during contact with a stiff environment, the values of the virtual mass and damping are increased. The proposed schemes are evaluated through simulations using a 2-DOF manipulator in MATLAB Simulink and Simscape. In addition, experiments were conducted by using a 6-DOF manipulator (Indy7, Neuromeka) and a six-axis force/torque sensor (RFT80-6A01, Robotous). The results of the simulation and experiment showed that the proposed control improves the performance while performing complicated tasks in stiff and soft environments.

This paper is organized as follows. The background of impedance control is explained in Sect. 2. Sect. 3 provides an overview of Hybrid Impedance and Admittance control. Sections 4 and 5 discuss the results of the simulations and experiments, and Sect. 6 presents the conclusions.

2 Background

2.1 Problem statement

For a multi-DOF manipulator, the dynamics model is expressed as

$$\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{G}(\mathbf{q}) + \boldsymbol{\tau}_f = \boldsymbol{\tau} + \boldsymbol{\tau}_{\text{ext}} \quad (1)$$

where $\mathbf{q}$, $\dot{\mathbf{q}}$, and $\ddot{\mathbf{q}}$ represent the vectors of the position, velocity, and acceleration vectors of the joints, respectively. $\mathbf{M}(\mathbf{q})$ is the inertia matrix, $\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})$ is the centrifugal and Coriolis matrix, $\mathbf{G}(\mathbf{q})$ represents the gravitational torque, $\boldsymbol{\tau}_f$ is the friction torque vector, $\boldsymbol{\tau}$ is the control torque vector, and $\boldsymbol{\tau}_{\text{ext}}$ indicates the external torque vector of the environment. $\boldsymbol{\tau}_{\text{ext}}$ can be calculated as follows:

$$\boldsymbol{\tau}_{\text{ext}} = \mathbf{J}^T \mathbf{F}_{\text{ext}} \quad (2)$$

where $\mathbf{J}$ denotes the Jacobian matrix.

To ensure that the manipulator performs the desired interaction, the manipulator should be controlled using the following dynamic equation in the task space:

$$\mathbf{M}_d(\ddot{\mathbf{x}} - \ddot{\mathbf{x}}_0) + \mathbf{D}_d(\dot{\mathbf{x}} - \dot{\mathbf{x}}_0) + \mathbf{K}_d(\mathbf{x} - \mathbf{x}_0) = \mathbf{F}_{\text{ext}} \quad (3)$$

where the $\mathbf{M}_d$, $\mathbf{D}_d$, and $\mathbf{K}_d$ denote the desired virtual inertia, damping, and stiffness, respectively. $\mathbf{x}$, $\dot{\mathbf{x}}$, and $\ddot{\mathbf{x}}$ denote the

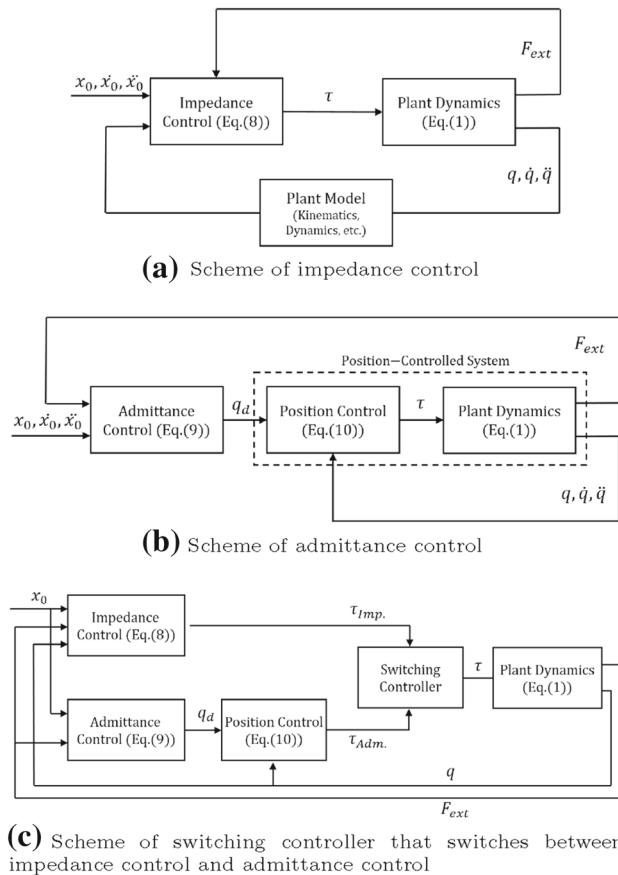


Fig. 1 Scheme of impedance controls

position, velocity, and acceleration vectors, respectively. $\mathbf{x}_0$ denotes the desired equilibrium position vector.

The errors of the desired equilibrium position and current position in the task and joint spaces are

$$\mathbf{e} = (\mathbf{x} - \mathbf{x}_0), \quad \mathbf{e}_\theta = (\mathbf{q} - \mathbf{q}_0) \quad (4)$$

The dynamic equation is rewritten as follows:

$$\mathbf{M}_d \ddot{\mathbf{e}} + \mathbf{D}_d \dot{\mathbf{e}} + \mathbf{K}_d \mathbf{e} = \mathbf{F}_{\text{ext}} \quad (5)$$

2.2 Impedance control

From equation (1), the control torque for Impedance Control is given by

$$\tau = \mathbf{M} \ddot{\mathbf{q}} + \mathbf{C} \dot{\mathbf{q}} + \mathbf{G} + \tau_f - \tau_{\text{ext}}. \quad (6)$$

From equations (3) and (4), the acceleration vector $\ddot{\mathbf{q}}$ can be formulated as

$$\begin{aligned} \ddot{\mathbf{q}} &= \mathbf{J}^{-1}(\ddot{\mathbf{x}} - \dot{\mathbf{J}}\dot{\mathbf{q}}) \\ &= \mathbf{J}^{-1}(\mathbf{M}_d^{-1}(\mathbf{F}_{\text{ext}} - \mathbf{D}_d \dot{\mathbf{e}} - \mathbf{K}_d \mathbf{e}_\theta) + \ddot{\mathbf{x}}_0 - \dot{\mathbf{J}}\dot{\mathbf{q}}). \end{aligned} \quad (7)$$

By substituting equation (7) into (6), we can derive the impedance control law as follows:

$$\begin{aligned} \tau_{\text{imp}} &= \mathbf{M} \mathbf{J}^{-1}(\mathbf{M}_d^{-1}(\mathbf{F}_{\text{ext}} - \mathbf{D}_d \dot{\mathbf{e}} - \mathbf{K}_d \mathbf{e}) + \ddot{\mathbf{x}}_0 - \dot{\mathbf{J}}\dot{\mathbf{q}}) \\ &\quad + \mathbf{C} \dot{\mathbf{q}} + \mathbf{G} + \tau_f - \tau_{\text{ext}} \end{aligned} \quad (8)$$

As shown in Fig. 1a, the impedance controller outputs the control torque required to perform a motion. To calculate the control torque, it is essential to have information regarding the kinematics and dynamics of the system. The plant is assumed to be an ideal force-controlled system in which the control torque can perfectly compensate for the disturbances. The external force $\mathbf{F}_{\text{ext}}$ and joint information were provided as feedback to the controller.

2.3 Admittance control

From the dynamic equation (3), the dynamics equation for the desired trajectory for admittance control can be written as

$$\mathbf{M}_d(\ddot{\mathbf{x}}_d - \ddot{\mathbf{x}}_0) + \mathbf{D}_d(\dot{\mathbf{x}}_d - \dot{\mathbf{x}}_0) + \mathbf{K}_d(\mathbf{x}_d - \mathbf{x}_0) = \mathbf{F}_{\text{ext}} \quad (9)$$

where $\mathbf{x}_d$, $\dot{\mathbf{x}}_d$, and $\ddot{\mathbf{x}}_d$ denote the desired position, velocity, and acceleration vectors of admittance control, respectively. The desired position in the work space is derived from this equation and can be transformed into the joint space as $\mathbf{q}_d$ through inverse kinematics.

To follow the desired trajectory $\mathbf{q} = \mathbf{q}_d$, a position-controlled system is implemented in the plant. For the position controller, the control torque is given by

$$\tau_{\text{adm}} = \mathbf{K}_p(\mathbf{q}_d - \mathbf{q}) - \mathbf{K}_v \dot{\mathbf{q}} + \mathbf{C} \dot{\mathbf{q}} + \mathbf{G} + \tau_f \quad (10)$$

where $\mathbf{K}_p$ and $\mathbf{K}_v$ denote the positive definite matrices for the control gain of the position controller.

As shown in Fig. 1b, the admittance controller receives the external force $\mathbf{F}_{\text{ext}}$ from the environment and outputs the desired position in the joint space to the position-controlled system. In the position controller, a PD controller was implemented in this study, and the control torque τ_{adm} was applied to the plant dynamics.

2.4 Unified impedance and admittance control

In a previous study, Ott et al. proposed a hybrid framework for a unified impedance and admittance control strategy, in which the controllers were switched rapidly [19,20]. From the hybrid framework, the control torque of the unified impedance and admittance control τ_{uni} can be determined as follows:

$$\tau_{\text{Uni}} = \begin{cases} \tau_{\text{Imp}} : t_0 + k\delta \leq t \leq t_0 + (k+1-n)\delta \\ \tau_{\text{Adm}} : t_0 + (k+1-n)\delta \leq t \leq t_0 + (k+1)\delta \end{cases} \quad (11)$$

where τ_{Imp} and τ_{Adm} are the control torques of impedance control and admittance control, respectively, t_0 is the initial time, δ is the switching period, k is an integer that has values of $0, 1, \dots, N$, and $n \in [0, 1]$ is the duty cycle. For example, if the value of n is 0, only the impedance control is applied, and if the value is 1, it means that only the admittance control is applied.

Figure 1c shows the scheme of the switching controller that defined by switching conditions δ and n . The switching conditions should be properly defined to optimize the performance in an environment with varying stiffness.

3 Hybrid impedance and admittance control

3.1 Motivation

As mentioned, admittance control exhibits good performance in soft environments, such as free motion. In contrast, impedance control exhibits a good performance in stiff environments. If the controller is switched according to the changes in the environment, stable interaction is possible in both soft and stiff environments. However, when the task requires interaction with both soft and stiff environments, the use of either admittance control or impedance control alone cannot be a solution. In some tasks, when the manipulator interacts with a stiff environment along a direction, it may be required to perform an interaction with a soft environment in another direction. For example, a deburring or mopping task requires interaction with both soft and stiff environments.

In the previous studies [20,21], in the unified impedance and Admittance control, the controller was adapted by changing the duty cycle n . For example, in the free motion, the duty cycle should be one $n = 1$, and in the contact motion, duty cycle should be zero $n = 0$ for the best performance. In the contact and sliding motions, the duty cycle for the best performance can be determined by variables such as the friction of contact, stiffness of the position controller, and sliding velocity. This can be a good controller in a general environment with varying stiffnesses. However, in general tasks, it is difficult to accurately determine the duty cycle.

In this section, we propose a novel control scheme that enables interaction with both soft and stiff environments by properly switching the controller when required. The switching conditions of the proposed control scheme are set for each hardware, and the manipulator can perform best in the free motion, contact motion, and contact and sliding motion without changing the parameters.

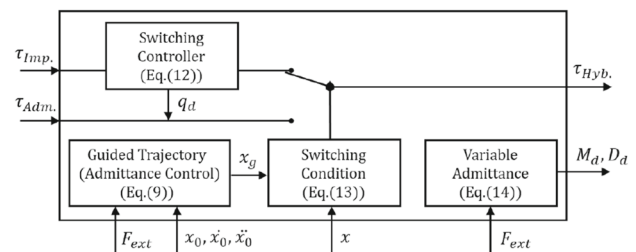


Fig. 2 Switching scheme of hybrid impedance and admittance control

The switching scheme of hybrid impedance and admittance control is described as shown in Fig. 2. The switching condition is determined using guided trajectory $\mathbf{x}_g$. Subsequently, the switching controller smoothly transitions impedance control to admittance control with a constant control torque. Moreover, a variable admittance was implemented to achieve stable interactions along the forced contact direction.

3.2 Switching condition

Based on the switching controller, the switching condition is the main contribution of this study. In the previous studies [20,21], the control was periodically switched. However, in the proposed hybrid impedance and admittance control scheme, impedance control is the basic control scheme, which is switched to admittance control whenever the switching condition is satisfied. To develop the switching condition, a guided trajectory $\mathbf{x}_g$ was introduced. When the manipulator is in impedance control or admittance control, the guided admittance control is also calculated independently for the guided trajectory. As shown in Fig. 2, the guided admittance control is calculated from the external force $\mathbf{F}_{\text{ext}}$ and the terms related to the desired equilibrium point, $\mathbf{x}_0$, $\dot{\mathbf{x}}_0$, and $\ddot{\mathbf{x}}_0$. This is a type of reference position, which is compared with the current position and determines the control method.

The switching condition for admittance control is shown below. The current position is compared with the guided position.

$$\begin{aligned} &\text{if } \|\mathbf{x}_g - \mathbf{x}\| > \alpha, \quad \text{Admittance Control} \\ &\text{else,} \quad \text{Impedance Control} \end{aligned} \quad (12)$$

where $\|\mathbf{x}_g - \mathbf{x}\|$ is the Euclidean distance between the guided position and the current position, indicating the position error in the task space. The newly adopted positive constant α indicates the acceptable position error. If the position error $\|\mathbf{x}_g - \mathbf{x}\|$ is larger than α , the system is controlled using the admittance controller to reduce the position error. Based on this basic scheme, additional strategies are available for interaction with both soft and stiff environments.

Based on the condition given by equation (12), during interaction with a soft environment, the manipulator is under impedance control with an accurate position within an acceptable error. By contrast, when the error exceeds the acceptable error, the manipulator is under admittance control. During the contact motion in a stiff environment, the manipulator is under impedance control with stable interaction until the position has an acceptable error. If the manipulator is in forced contact and needs to move along another vertical direction, impedance control is suitable for forced contact and admittance control is suitable for motion along the other vertical direction. To achieve interaction in both directions, the manipulator under admittance control should have a stable interaction with the forced contact by properly changing the parameters of admittance control.

When the controller is switched from impedance control to admittance control, two additional states variables are required, namely, $\mathbf{q}_d$ and $\dot{\mathbf{q}}_d$, which denote the desired position and velocity in the joint space, respectively. By contrast, when the controller is switched from admittance control to impedance control, no additional states are required.

From equation (10), which provides the control torque of the admittance control, the following equation is obtained:

$$\mathbf{q}_d = \mathbf{q} + \mathbf{K}_p^{-1}(\tau_{\text{Imp}} + \mathbf{K}_v \dot{\mathbf{q}} - \mathbf{C}\dot{\mathbf{q}} - \mathbf{G} - \tau_f) \quad (13)$$

where $\dot{\mathbf{q}}_d$ denotes the derivative of $\mathbf{q}_d$.

Whenever switching from impedance to admittance control, $\mathbf{q}_d$ and $\dot{\mathbf{q}}_d$ are initialized through Equation (13) to generate a continuous control torque.

3.3 Switched system stability

To analyze the stability of the switching system, the subsystem models and the environment were based on a previous study [20], which has a single degree-of-freedom system without friction, model uncertainty, and time delay. The switched system can be described as:

$$\dot{\mathbf{x}}_\sigma = \begin{cases} \mathbf{A}_i \mathbf{x}_i & \text{if } \|\mathbf{x}_g - \mathbf{x}\| \leq \alpha \\ \mathbf{A}_a \mathbf{x}_a & \text{if } \|\mathbf{x}_g - \mathbf{x}\| > \alpha \end{cases} \quad (14)$$

where σ is the switching signal with $\sigma \in \{i, a\}$, $\mathbf{x}_i = [e \ \dot{e}]^T$, $\mathbf{x}_a = [e \ \dot{e} \ \ddot{e}]^T$, $e = x_0 - x$, $\dot{e} = \dot{x}_0 - \dot{x}$, $\ddot{e} = \ddot{x}_0 - \ddot{x}$,

$$\mathbf{A}_i = \begin{bmatrix} 0 & 1 \\ -(K_d + k_e)/M_d & -D_d/M_d \end{bmatrix} \quad (15)$$

and

$$\mathbf{A}_a = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -(k_p + k_e)/m & -k_v/m & k_p/m & 0 \\ 0 & 0 & 0 & 1 \\ -k_e/M_d & 0 & -k_d/M_d & -D_d/M_d \end{bmatrix}. \quad (16)$$

Because each subsystem has a different dimensional state vector, a multiple Lyapunov method was used [22]. Assuming that each subsystem is a continuous-time system, the impedance control is a Hurwitz matrix because $\mathbf{M}_d, \mathbf{D}_d$ is positive definite. Thus, the subsystem is asymptotically stable. On the other hand, in the case of admittance control, the stiffness of the environment affects the stability, so it cannot always guarantee the stability. A study has been conducted on the multiple Lyapunov function approach for a switched system when the subsystem is unstable [23].

A switched system with different state vectors is expressed as:

$$\dot{\mathbf{x}}_a = f_a(\mathbf{x}_a(t)) \quad t \in \omega(S|a) \quad (17)$$

where $\mathbf{x}_a \in \mathbb{R}^{n_a}$ and $\omega(S|a)$ is the set of time intervals during which subsystem a is active. S is the switching sequence related to the switched system.

Considering impedance control ($\sigma = i$), the derivative of the Lyapunov function satisfies following equation.

$$\dot{V}_i(\mathbf{x}_i(t)) < 0, \quad t \in \Omega(S|i). \quad (18)$$

Thus, the impedance control satisfies condition.1 in Theorem.1[23]. In the case of admittance control, the rate of change of Lyapunov may increase depending on the stiffness of the environment. Therefore, in this controller, we assumed that the rate of change of the Lyapunov function, which is the worst case, increases. For the subsystem to satisfy Condition 2 in Theorem 1, it must remain as an impedance controller in the last state of the system. Alternatively, the impedance control must be a switching system before switching to the admittance control. The selection of α in the next section plays an important role in satisfying this condition. Because admittance control is always switched to the impedance control according to the designed switching condition, the two conditions defined in Theorem 1 are satisfied [23]. Therefore, the proposed system guarantees asymptotic stability even if admittance control is unstable.

3.4 Boundary of switching condition value (α)

It is proven that the proposed system is the asymptotically stable. Furthermore, by considering the impulse effect that occurs during switching, it can be referred to in selecting the α in the switching condition [22]. The impulse effect is caused

by the state vector added when switching to the admittance control, and the related Lyapunov candidate function is as follows:

$$V_a(\mathbf{x}_a) = \frac{1}{2} \{m\dot{x}^2 + m_d\dot{e}_d^2 + k_d e_d^2 + k_p(e - e_d)^2\}. \quad (19)$$

The constants, m , m_d , k_d , k_p , are the same value in the equation (16) and are composed of positive values. It can also be interpreted as a sum of kinetic energy and potential energy for the admittance control.

In the case of the guide position, it can be seen as the same as x_d ; therefore, the following relationship can be derived:

$$x_g - x \Rightarrow x_d - x \Rightarrow e_d - e = \alpha. \quad (20)$$

Therefore, α is related to the equation (19), and it can limit the impulse effect to a low level by minimizing the value. The minimum value in the boundary of the α value is the maximum value of the cartesian position error by the internal position controller. This is because when the switching condition of the controller is smaller than the position error, there is a possibility that the admittance control status is maintained for most of the time in the finite control time. According to the switching theory [23], this is an unstable system because it does not change to a stable subsystem, which means impedance control in a steady state. Therefore, the α should be determined within the range in which admittance control does not work in the steady state of the system. The simulation results related to α are presented in the next section.

3.5 Variable admittance

Previous research has shown that the larger the virtual mass and damper, the greater is the stability of the system [14, 25]. If the virtual mass and damper are increased only in the direction in which the force is applied, the system will be stable in the direction of the stiff environment and will have initial admittance parameters in the other vertical direction. If the x axis corresponds to the force direction, the virtual admittance parameters should be changed as follows:

$$\mathbf{M}_d = \begin{bmatrix} w_m m_d & 0 & 0 \\ 0 & m_d & 0 \\ 0 & 0 & m_d \end{bmatrix}, \mathbf{D}_d = \begin{bmatrix} w_d d_d & 0 & 0 \\ 0 & d_d & 0 \\ 0 & 0 & d_d \end{bmatrix}. \quad (21)$$

where $\mathbf{M}_d \in \mathbb{R}^{3 \times 3}$ and $\mathbf{D}_d \in \mathbb{R}^{3 \times 3}$ denote the virtual mass and damping matrix in the task space, respectively, and w_m and w_d represent the parameters that change the admittance. By changing the admittance parameters along the forced contact direction, stable interaction was achieved while retaining good performance for the sliding motion in the other vertical direction.

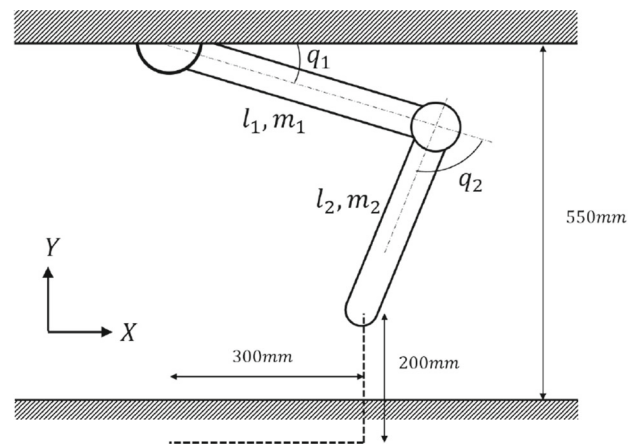


Fig. 3 2-DOF manipulator with the target trajectory

4 Simulation

4.1 Simulation environment

Simulations were conducted using a two-DOF manipulator to verify the proposed control strategy. A 2-DOF manipulator was built and simulated using MATLAB Simulink and Simscape.

As shown in Fig. 3, the 2-DOF manipulator is based on the example presented in Sect. 2. For the simulation, the following parameters were newly defined: mass of links (m_1, m_2), model mass error ($\hat{m}_1, \hat{m}_2$), length of links (l_1, l_2), viscous and Coulomb frictions of the joints ($\tau_{fv1}, \tau_{fc1}, \tau_{fv2}, \tau_{fc2}$), parameters of contact surface (K_E, D_E, F_{fv}, F_{fc}), impedance parameters ($\mathbf{M}_d, \mathbf{K}_d, \mathbf{D}_d$), and PD gain of the position controller ($\mathbf{K}_p, \mathbf{K}_v$). Table 1 lists the values of the parameters used in the simulations.

4.2 Simulation task

The simulation task is illustrated in Fig. 3. The dashed line shows the target trajectory for the manipulator. From the home position $\mathbf{q}_0 = [-10^\circ - 90^\circ]^T$, the target position moves 200 mm along the $-Y$ axis for 4 s. During this motion, the end of the manipulator contacts with the bottom plane and starts to apply a force. When the force is applied to the bottom plane, the magnitude of the force is approximately 35 N. After a pause for 1 s, the target position moves 300 mm along the $-X$ axis for 3 s. In the simulation, the tasks included three types of motions: free motion, contact motion, and forced/sliding motion. These motions represent the general interaction between the robot manipulator and the environment. Free motion represents interaction with a soft environment, and contact motion represents interaction with a hard environment. The forced/sliding motion represents interactions with a complex environment.

Table 1 Simulation parameters for the 2-DOF manipulator

Real mass of links	m_1, m_2	1kg
Model mass of links	$\hat{m}_1, \hat{m}_2$	0.8kg
Length of 1st link	l_1	0.5m
Length of 2nd link	l_2	0.35m
Viscous friction of joints	τ_{fv1}, τ_{fv2}	0.5Nm
Coulomb friction of joints	τ_{fc1}, τ_{fc2}	0.5Nms/rad
Stiffness of environment	K_E	10^6 N/m
Damper of environment	D_E	10^4 Ns/m
Viscous friction of contact	F_{fv}	0.01N
Coulomb friction of contact	F_{fc}	0.01Ns/N
Mass of impedance control	$\mathbf{M}_d$	31kg
Stiffness of impedance control	$\mathbf{K}_d$	3001N/m
Damper of impedance control	$\mathbf{D}_d$	$1.4\sqrt{m_d k_d} \text{ Ns/m}$
P gain of position controller	$\mathbf{K}_p$	$\begin{bmatrix} 10^4 & 0 \\ 0 & 3 \times 10^3 \end{bmatrix}$
D gain of position controller	$\mathbf{K}_v$	$\begin{bmatrix} 50 & 0 \\ 0 & 12 \end{bmatrix}$

Simulations were performed to compare four control methods: the two basic control methods, namely impedance control and admittance control, and the two switching control methods, namely the unified impedance and admittance control and the proposed hybrid impedance and admittance control. For unified impedance and admittance control, the control gains were $\delta = 20$ and $n = 0.5$. For the hybrid impedance and admittance control, the control gains were selected practically as $\alpha = 5 \text{ mm}$, $w_m = 1$, and $w_d = 5$.

4.3 Simulation result

Figure 4a shows the contact forces obtained using the four control algorithms. The dashed line represents the reference contact force, and the straight line represents the result of each control algorithm. In the graphs, the tasks were divided into three states, and each control had a different response. With the admittance control, free motion showed the most accurate result; however, the contact was unstable. With the impedance control, the contact was stable; however, the most inaccurate results were obtained for the force and position errors in all states. With the unified impedance and admittance control, the values of force and position accuracy were midway between those obtained with impedance control and admittance control. The stability was quite good, but there was a small vibration during the contact state because of the switching controller. With hybrid impedance and admittance control, both the force and position accuracy showed good results. However, during the contact/sliding motion state, periodic vibrations occurred because of the switching controller.

Regarding the position accuracy, the error between the reference position and the controlled positions is shown in Fig. 4b. During the simulation task, the admittance and hybrid controls had a small error. The impedance control had the largest error, and the unified control had a moderate value. The results are summarized as follows:

- Contact Stability:
Impedance > Hybrid > Unified > Admittance
- Position Accuracy:
Admittance > Hybrid > Unified > Impedance
- Force Accuracy:
Hybrid > Unified > Impedance > Admittance

For a more accurate comparison, the performances of the unified impedance and admittance control were also compared according to the change in the control gains. As shown in Fig. 5a, when the switching period δ increases, the accuracy increases, but the vibration also increases owing to the influence of the admittance control. The results are similar to those shown in Fig. 5b because when the duty cycle n increases, the influence of admittance control increases. For the overall performance, the control gains were defined as $\delta = 20$ and $n = 0.5$ in this simulation.

The advantages of the hybrid impedance and admittance control schemes are illustrated in Fig. 6. The contact force and switch index were compared for various acceptable position errors α . For the switching index, 0 indicates impedance control and 1 indicates admittance control.

When the acceptable position error is small, such as $\alpha = 1 \text{ mm}$, admittance control is mainly used during the simulation, and the contact stability is lower than that of the

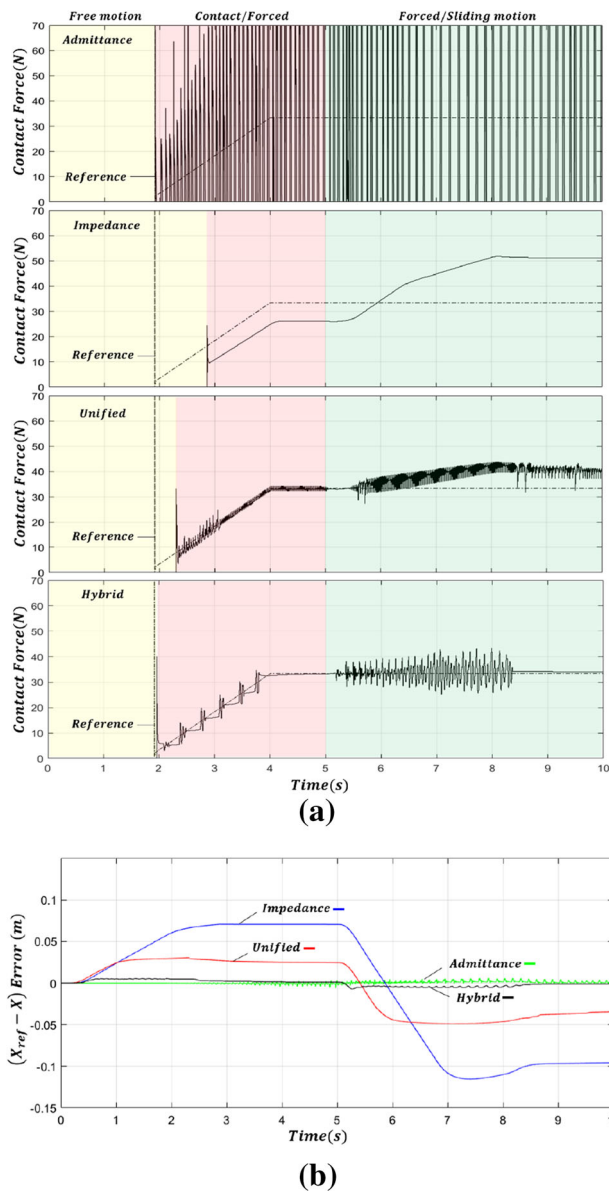


Fig. 4 Time history of **a** contact force **b** position error along X -axis with four control algorithms for the simulation task

others. When $\alpha = 50 \text{ mm}$, impedance control is mainly used, and the force/position accuracy and contact stability are both lower than the others. When the acceptable position error has a proper value of $\alpha = 5 \text{ mm}$, the force/position accuracy and contact stability have the best results.

During free motion, admittance control is primarily used to follow the target position. In contrast, during contact/forced motion, impedance control is mainly used for stable interactions. Finally, during the forced/sliding motion, the control is switched between impedance control and admittance control to follow the target position and perform stable interactions. Thus, the proposed control strategy showed good performance with regard to both the contact

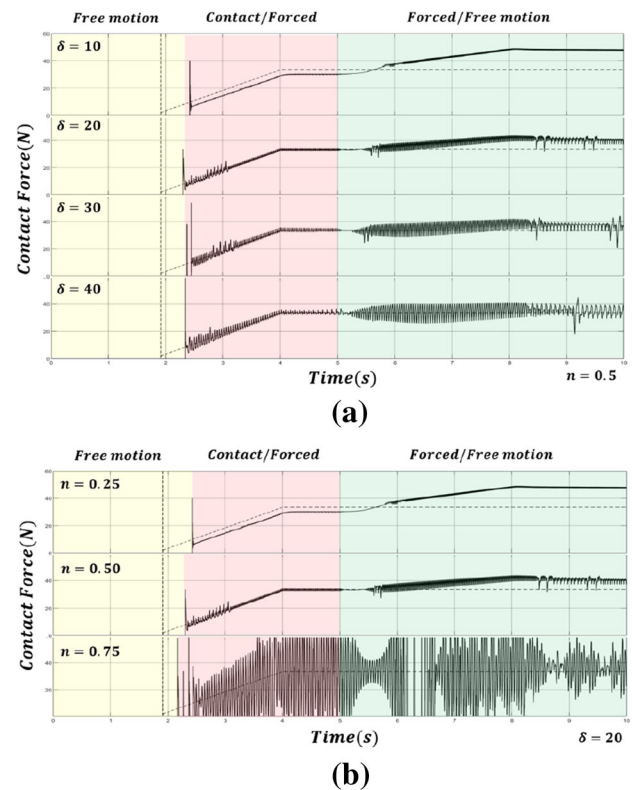


Fig. 5 Result of unified impedance and admittance control with various control gains δ and n : **a** various δ with $n = 0.5$ and **b** various n with $\delta = 20$

stability and force/position accuracy when the acceptable position error was set properly.

5 Experiment

5.1 Experimental setup

As shown in Fig. 7, the experimental system consisted of a 6-DOF industrial manipulator (Indy7, Neuromeka) and a six-axis force/torque sensor (RFT80-6A01, Robotous). The force/torque sensor was attached to the end-effector of the manipulator. In the experiment, the external force $\mathbf{F}_{\text{ext}}$ was measured using a force/torque sensor. The manipulator was controlled using a real-time operating system called Xenomai (based on Linux) at a frequency of 1 kHz , and the sensor data were obtained at 1 kHz with a low-pass filter for noise reduction.

Impedance control and admittance control were implemented on the robot based on the kinematics and dynamics of the robot hardware. Gravity and friction compensation were also implemented on the robot. The unified impedance and admittance control and the hybrid impedance and admittance control strategies were implemented based on the basic

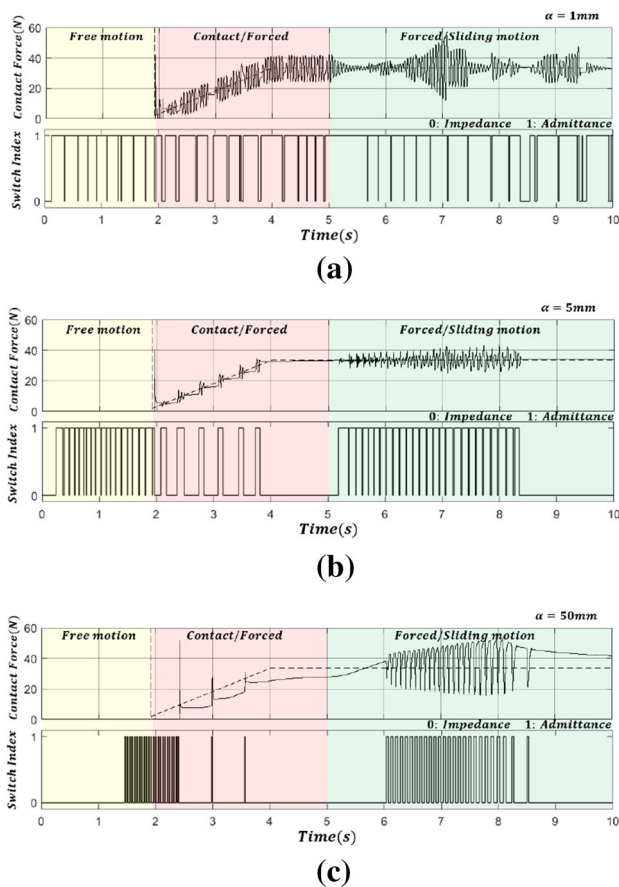


Fig. 6 Result of hybrid impedance and admittance control with various acceptable position error α : **a** $\alpha = 1 \text{ mm}$, **b** $\alpha = 5 \text{ mm}$, and **c** $\alpha = 50 \text{ mm}$

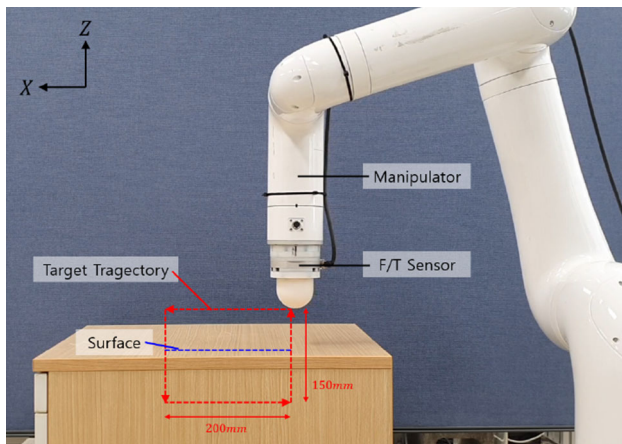


Fig. 7 Experimental setup and task

impedance control. Data on the contact force and Cartesian position were obtained and compared.

Table 2 Experiment parameters for 6-DOF manipulator

Desired mass of impedance control	M_d	$\begin{bmatrix} 10I & 0 \\ 0 & 5I \end{bmatrix}$
Desired stiffness of impedance control	K_d	$\begin{bmatrix} 500I & 0 \\ 0 & 500I \end{bmatrix}$
Desired damper of impedance control	D_d	$\begin{bmatrix} 200I & 0 \\ 0 & 200I \end{bmatrix}$
P gain of position controller	K_p	$\begin{bmatrix} 10^4I & 0 \\ 0 & 10^4I \end{bmatrix}$
D gain of position controller	K_v	$\begin{bmatrix} 400 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1200 & 0 & 0 & 0 & 0 \\ 0 & 0 & 600 & 0 & 0 & 0 \\ 0 & 0 & 0 & 300 & 0 & 0 \\ 0 & 0 & 0 & 0 & 300 & 0 \\ 0 & 0 & 0 & 0 & 0 & 300 \end{bmatrix}$

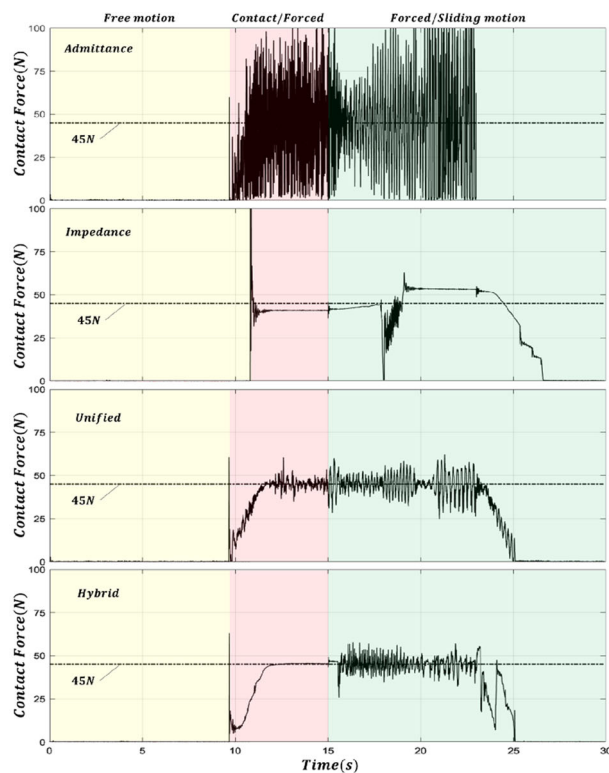
5.2 Experimental task

The experimental task is shown in Fig. 7. The dashed red line represents the target trajectory of the manipulator. From the home position, the target position moved 200 mm along the X axis in free space. Subsequently, the target position moved 150 mm along the $-Z$ axis. During the motion, the end of the manipulator contacted the desk plane and started applying a force. When the force was applied to the bottom plane, the magnitude of the force was approximately 45 N. After a pause of 2 s, the target position moved 200 mm along the $-X$ axis direction. Again, after a pause of another 2 s, the target position moved 150 mm along the Z axis to the home position.

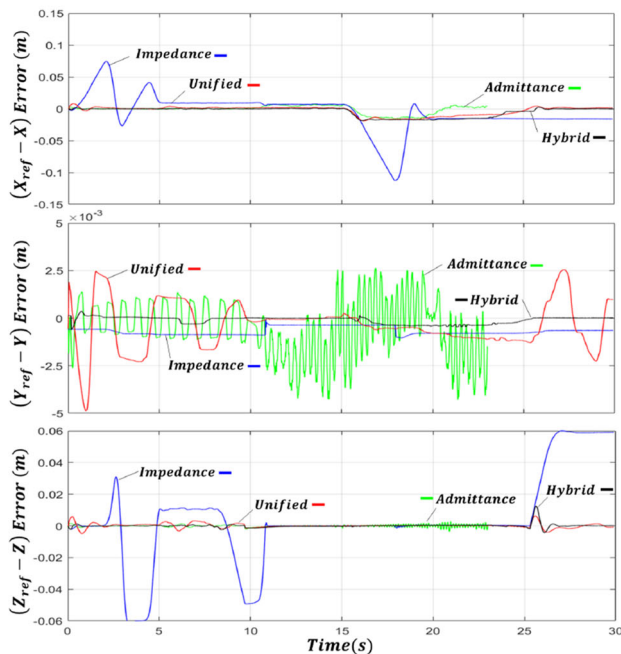
Similarly, in the simulations, the experiments were performed using the four control methods. For unified impedance and admittance control, the control gains were $\delta = 10 \text{ ms}$ and $n = 0.5$. For the hybrid impedance and admittance control, the control gains were selected practically as $\alpha = 5 \text{ mm}$, $w_m = 2$, and $w_d = 10$.

5.3 Experimental results

The experimental task was similar to the simulation task, and the results were similar. Figure 8a shows the plots of contact force versus time obtained using the four control algorithms. The expected contact force was 45 N at a position between the target position and the surface position. The dashed line represents the expected contact force, and the straight red line shows the results of each control scheme. Figure 8b shows the position accuracy, error between the reference position, and controlled position on each axis.



(a)



(b)

Fig. 8 Time history of **a** contact force and **b** position error with four control algorithms for the experimental task

Table 3 RMS errors of position and force

Control	Position RMS error [m]	Force RMS error [N]
Admittance	0.0065	29.3046
Impedance	0.0397	8.8497
Unified	0.0076	5.5066
Hybrid	0.0067	4.1651

For a better evaluation of the performance, the RMS errors of the position and force are shown in Table 3. The RMS measurement range of the position error was the area in which each controller operated. On the other hand, the RMS measurement range of the force was set from the start of the green region, where the desired force was maintained (Fig. 8a).

While performing the task, the unified and hybrid control strategies exhibited small RMS errors. In the case of impedance control, there was a large error in position, and admittance control had a large error in force.

Unlike that in the simulation results, admittance control was stopped because of hardware limitations. Impedance control had the largest initial contact force because the free motion was not smooth and it hit the plane strongly. After contact, the contact force remained stable, but the vertical motion became worse than that in the simulation. With unified impedance and admittance control, the force and position accuracy were quite good, but there were small vibrations during the contact and sliding motions. With hybrid impedance and admittance control, both the force and position accuracy showed good results.

Thus, both the unified and hybrid control strategies showed good performance with regard to contact stability and force/position accuracy, as shown in Table 3. However, unified control always causes vibrations because of the periodic switching controller. In contrast, hybrid control switches between controllers according to the situation.

6 Conclusion

This study introduced a hybrid control strategy with both impedance control and admittance control. In a conventional robot manipulator, impedance control exhibits poor performance with regard to the position accuracy. To achieve better position accuracy, impedance control should be stiff. However, it has limitations with regard to its low impedance response. The proposed control scheme switches the controller from impedance control to admittance control whenever high accuracy is required. The proposed control scheme enables smoother robot movement under any impedance condition.

The switching condition is an important contributor to the control scheme. In this study, a guided trajectory $\mathbf{x}_g$ was introduced and used for comparison with the current position $\mathbf{x}$ to determine the switching condition. The effectiveness of the proposed control method was verified through simulations and experiments. A task involving free motion, contact motion, and forced/sliding motion was performed in both simulations and experiments. The results showed that the proposed hybrid control strategy performed better with respect to the position/force accuracy and contact stability. Furthermore, by changing the parameters α , w_m , and w_d , the proposed hybrid strategy can be implemented on various hardware platforms. However, the parameters α , w_m , and w_d should be set properly to achieve stable performance, which may still have some limitations.

This study is an extension of previous studies on stable-switching controllers [20,21]. Previous studies considered only the contact situation, whereas we additionally considered the contact transition from free to contact motion, which has environments with different stiffnesses and trajectories following contact. It was also confirmed that an impulsive force is generated when transitioning to another environment, but stable performance was demonstrated with the proposed controller. Furthermore, the previous controller must change the parameter n properly during the tasks. However, the parameters of the proposed controller depend on the hardware and do not need to be changed during the same tasks.

Based on the proposed concept of switched control system, a new switching condition can be introduced in future works. With a better switching condition with stable parameters, the performance of the hybrid impedance and admittance control system can be improved.

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